

Proton Radius Puzzle and Lattice QCD

Seminar II – Nuclear and Particle Physics

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Outline

The Proton and Its Significance

Measuring the Proton Radius

Lattice QCD

Gauge Ensembles at Mainz

Computing Form Factors on the Lattice

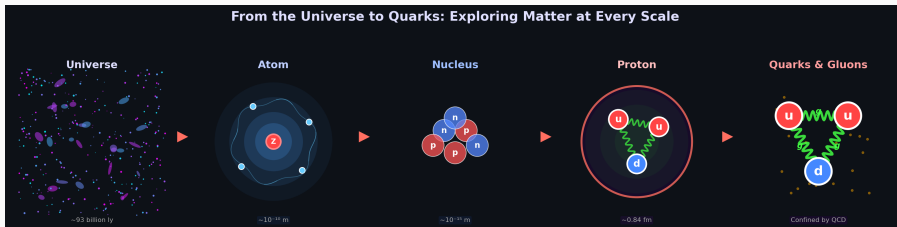
Results

Summary and Outlook

The Proton and Its Significance

Motivation: From the Universe to the Proton

The central goal of physics: understand the matter of the universe and the laws governing its motion.



- Ordinary visible matter contains quarks and leptons interacting through gauge fields
- The **proton**: quarks + gluons — How big is it?

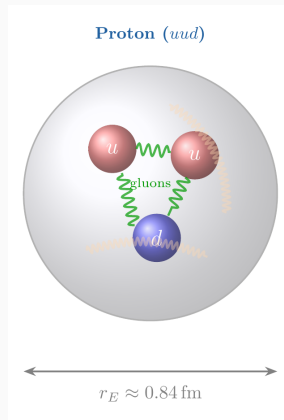
Hofstadter, Rev. Mod. Phys. **28**, 214 (1956) — Nobel Prize 1961.

Why Is the Proton So Important?

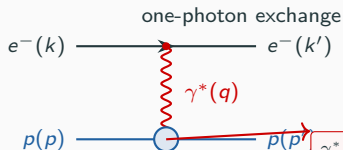
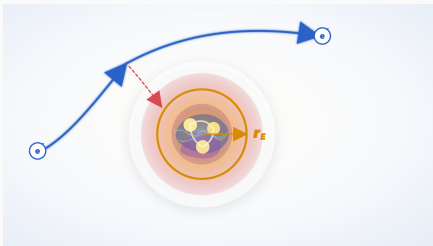
Why proton?

1. **Ordinary matter**: lightest baryon
2. **Internal structure**: charge, mass, spin
3. **QCD benchmark**: structure from first principles

Today: the charge radius r_E^p .



Measuring the Proton Radius



γ^* probes EM current
 $J_{\text{em}}^\mu = \sum_q e_q \bar{q} \gamma^\mu q$
 Breit frame: $J_{\text{em}}^0 \Rightarrow$
 $G_E(Q^2) \Rightarrow r_E$

1. Measure the differential cross section

$$\frac{d\sigma}{d\Omega} = \sigma_{\text{Mott}} S_p(Q^2, \theta).$$

σ_{Mott} is the point-charge QED baseline; S_p contains the proton structure through G_E and G_M .

2. Read the electric response at low Q^2

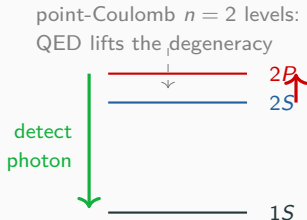
$$\begin{aligned}
 G_E(Q^2) &\simeq \int d^3r \rho_E(\vec{r}) e^{i\vec{q}\cdot\vec{r}} \\
 &= 1 - \frac{Q^2}{6} \langle r_E^2 \rangle + \mathcal{O}(Q^4).
 \end{aligned}$$

3. The initial slope defines the charge radius

$$\langle r_E^2 \rangle = -6 \left. \frac{dG_E}{dQ^2} \right|_{Q^2=0}.$$

Method 2: Lamb Shift Spectroscopy

Backup



$$\Delta E_{\text{res}} \equiv E(2P) - E(2S)$$

The laser measures ΔE_{res} . Known fine- and hyperfine-structure terms are removed to isolate the Lamb shift.

Pohl et al., Nature **466**, 213 (2010); Antognini et al., Science **339**, 417 (2013).

How is the resonance detected?



1. Prepare long-lived $2S$ muonic hydrogen.
2. Scan the $2S \rightarrow 2P$ laser frequency.
3. Detect the prompt $2P \rightarrow 1S$ photon.

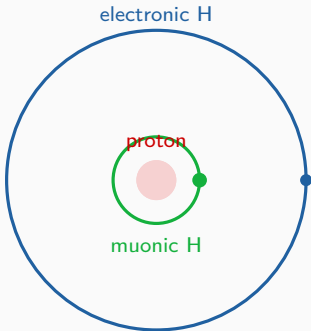
Measured resonance (2010)

$$\Delta E_{\text{res}} = 206.2949(32) \text{ meV},$$

$$r_E^P = 0.84184(67) \text{ fm}.$$

Why Muonic Hydrogen Is So Sensitive

Backup



Muon orbit ~ 200 times smaller

Finite proton size shifts an S -state

$$\Delta E_{\text{fs}}(nS) \propto |\psi_{nS}(0)|^2 r_E^2 \propto m_r^3 r_E^2.$$

- S -waves reach the proton: $|\psi_{nS}(0)|^2 > 0$.
- P -waves vanish at the origin and provide a nearby reference.
- Since $m_\mu/m_e \simeq 207$, the finite-size sensitivity is enhanced by roughly 10^7 .

Pohl et al., Nature **466**, 213 (2010); Antognini et al., Science **339**, 417 (2013).

Proton Radius Puzzle (2010–2026)

Backup

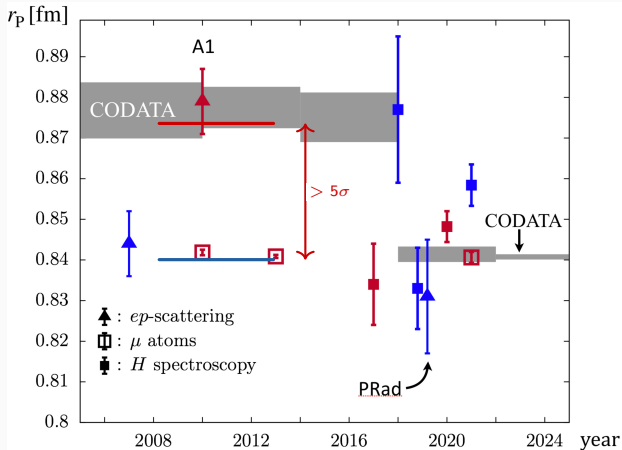


Figure message

Old split:

μH : 0.84184(67) fm,

CODATA-2010 : 0.8775(51) fm.

Same proton, different probes:

$> 5\sigma$.

This was the puzzle.

Modern checks:

PDG 2024: $r_E^p = 0.8409(4)$ fm;

H 2S–6P (2026): 0.8406(15) fm.

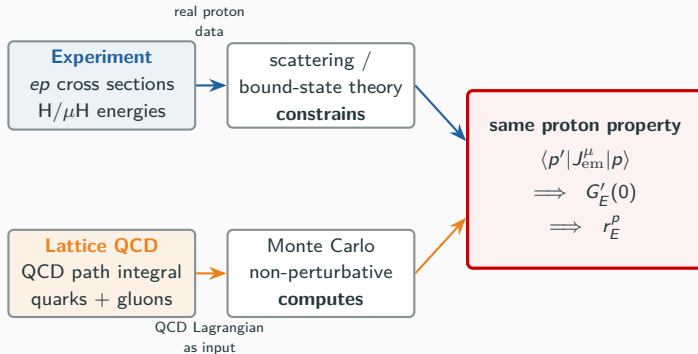
Why puzzle?

$$\langle r_E^2 \rangle = -6 G_E'(0)$$

Same proton current $J_{em}^\mu \Rightarrow$ same $G_E'(0)$: a proton property.

Lattice QCD

After experiments: can QCD itself predict the same proton response?



Message: experiments infer from data; lattice QCD computes from QCD; both use the same radius definition.

A QCD observable is an average over fields

$$\langle O \rangle = \frac{1}{Z_E} \int [D\Phi] O[\Phi] e^{-S_E[\Phi]}, \quad \Phi = (U, \bar{\psi}, \psi).$$

$$t \rightarrow -i\tau : \quad e^{iS_M[\Phi]} \longrightarrow e^{-S_E[\Phi]}$$

Why Euclidean?

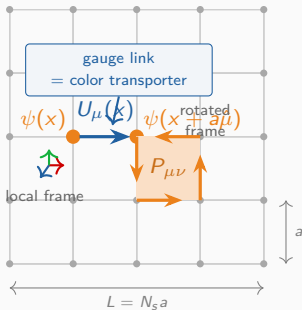
- e^{iS_M} is oscillatory.
- e^{-S_E} behaves like a statistical weight.
- The action tells us which field configurations are likely.

Monte Carlo average

$$\langle f \rangle = \int dx f(x) P(x) \simeq \frac{1}{N} \sum_i f(x_i).$$

Draw likely configurations
 $x_i \sim P(x)$, measure $f(x_i)$,
average.

On the lattice: $U_i \sim P[U]$, then $\langle O \rangle \simeq N_{\text{cfg}}^{-1} \sum_i O[U_i]$, with $O = C_2$ or C_3^μ .



site: quark field $\psi(x)$ link: transporter $U_\mu(x)$

Quarks live on sites

$$\psi(x) \rightarrow \Omega(x)\psi(x)$$

$\psi(x)$ is a colour vector at one lattice point.

Gluons live on links

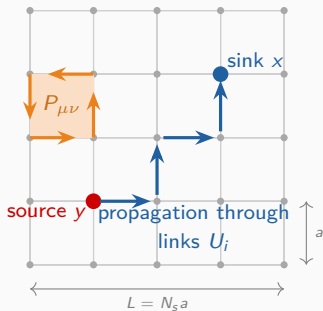
$$U_\mu(x) = \mathcal{P}e^{ig \int_x^{x+a\hat{\mu}} A_\mu(z) dz^\mu}$$

U_μ parallel-transport colour from x to $x + a\hat{\mu}$.

$\bar{\psi}(x) U_\mu(x) \psi(x + a\hat{\mu})$ is gauge invariant.

Plaquette A plaquette $P_{\mu\nu}$ is the smallest closed link loop; summing plaquettes gives $S_g[U]$, with $a \rightarrow 0$ recovering $\text{Tr } F_{\mu\nu} F_{\mu\nu}$ in Euclidean space.

Wilson, PRD **10**, 2445 (1974); Wittig, §5.2.2.



quark propagator = response from y to x

1. Fix one gauge background U_i

All links are known, so the lattice Dirac operator $D_q[U_i; m_q]$ is known.

2. Solve for the quark propagator

$$\sum_z D_q(x, z; U_i) S_q(z, y; U_i) = \delta_{xy}, \quad S_q = D_q^{-1}.$$

point source at $y \rightarrow$ response at x .

This response is the quark Green's function.

3. Build proton observables

The proton operator contracts three propagators (S_u, S_d, S_u) into C_2 and C_3^μ ; finally average over all U_i .

$$U_i \implies D_q[U_i] \implies S_q = D_q^{-1} \implies C_2, C_3^\mu.$$

Why Euclidean Time? Filtering the Proton

Backup

Proton operator: a uud source/sink

$$\chi_p^\alpha(x) = \epsilon^{abc} (u^{aT}(x) C \gamma_5 d^b(x)) u^{c\alpha}(x).$$

χ_p has proton quantum numbers and overlaps with proton/excited states.

Two-point correlator

$$C_2(t) = \sum_{\vec{x}} \langle \chi_p(\vec{x}, t) \bar{\chi}_p(\vec{0}, 0) \rangle.$$



Euclidean time acts as a filter

$$\begin{aligned} C_2(t) &= \langle 0 | \chi_p e^{-Ht} \bar{\chi}_p | 0 \rangle \\ &= \sum_n |A_n|^2 e^{-E_n t} \xrightarrow{t \text{ large}} |A_0|^2 e^{-M_N t}. \end{aligned}$$

$\bar{\chi}_p | 0 \rangle = \sum_n A_n | n \rangle$: not an energy eigenstate.

Large t keeps the lowest state

$$E_0 = M_N.$$

Large t : excited states die out; the slope of $\log C_2(t)$ gives M_N .

Insert the electromagnetic current

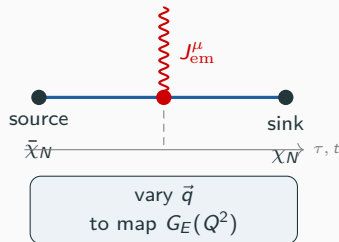
$$C_3^\mu(t, \tau, \vec{q}) = \langle \chi_N(t) J_{\text{em}}^\mu(\tau, \vec{q}) \bar{\chi}_N(0) \rangle.$$

$$0 \ll \tau \ll t \implies C_3^\mu|_{\text{ground state}} \propto \langle p' | J_{\text{em}}^\mu | p \rangle.$$

This is the lattice version of a virtual scattering experiment: create a proton, insert the current, destroy the proton.

Then decompose

$$\langle p' | J_{\text{em}}^\mu | p \rangle \rightarrow F_1(Q^2), F_2(Q^2) \rightarrow G_E(Q^2), G_M(Q^2).$$



Sachs, Phys. Rev. **126**, 2256 (1962); Djukanovic et al., PRL **132**, 211901 (2024).

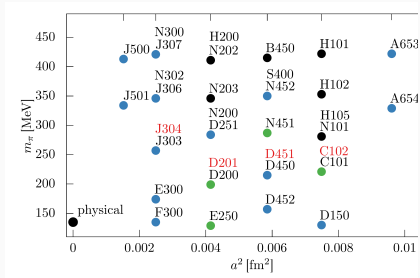
Gauge Ensembles at Mainz

Why Many Lattice Ensembles?

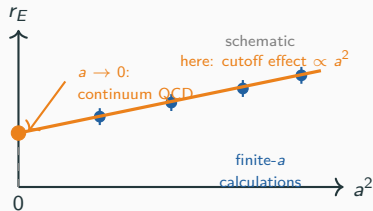
Backup

One ensemble is one finite lattice. The physical result requires several lattice spacings.

RG: change the cutoff while keeping the physics fixed



CLS varies a , m_π , and the box size L .



Changing a changes the cutoff $\Lambda_{UV} \sim 1/a$. Repeat the same observable at several matched a , then take the intercept at $a = 0$.

Bruno et al., JHEP **02**, 043 (2015); Bali et al., PRD **94**, 074501 (2016).

Computing Form Factors on the Lattice

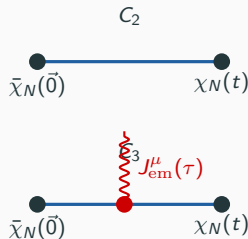
Goal: compute $G_E(Q^2)$ and $G_M(Q^2)$ from lattice QCD.

Two-point function (χ_N : nucleon interpolating operator):

$$C_2(\vec{p}, t) = \sum_{\vec{x}} e^{-i\vec{p}\cdot\vec{x}} \langle \chi_N(\vec{x}, t) \bar{\chi}_N(\vec{0}, 0) \rangle$$
$$\xrightarrow{t \text{ large}} |\mathcal{A}_0|^2 e^{-E_{\vec{p}} t}.$$

Three-point function (nucleon–current–nucleon, with $\vec{p}' = 0$):

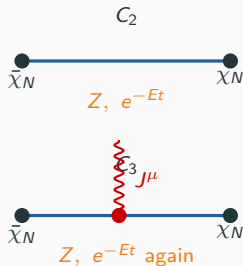
$$C_3^\mu(\vec{q}, t, \tau) = \sum_{\vec{x}, \vec{y}} e^{i\vec{q}\cdot\vec{y}}$$
$$\times \langle \chi_N(\vec{x}, t) J_{\text{em}}^\mu(\vec{y}, \tau) \bar{\chi}_N(\vec{0}, 0) \rangle.$$



C_2 : proton propagation. C_3 : insert J_{em}^μ and measure the response.

Djukanovic et al., PRL **132**, 211901 (2024) [arXiv:2309.07491].

Large-time behaviour



$$C_2(t, \vec{p}) \rightarrow |Z_p|^2 e^{-E_p t}.$$

$$C_3^\mu(t, \tau) \rightarrow Z_{p'} Z_p^* e^{-E_{p'}(t-\tau)} e^{-E_p \tau} \langle p' | J_{\text{em}}^\mu | p \rangle.$$

$$R^\mu \sim \frac{C_3^\mu}{C_2 \text{ combinations}} \rightarrow \text{known kinematics} \times \langle p' | J_{\text{em}}^\mu | p \rangle.$$

Overlap and propagation factors cancel; the current matrix element remains.

Maiani et al., NPB **293**, 420 (1987); Capitani et al., PRD **86**, 074502 (2012).

Core idea: the ratio gives a plateau Π_μ ; kinematics turns it into G_E and G_M .

Ground-state matrix element

$$R_\mu(\vec{q}, t, \tau) \xrightarrow{0 \ll \tau \ll t} \Pi_\mu(\vec{q}).$$

Plateau = a projected ground-state matrix element, with known kinematic factors.

Practical tension: ground-state isolation vs. noise.

Kinematic projection

$$\Pi_\mu(\vec{q}) = K_\mu^E(\vec{q}) G_E(Q^2) + K_\mu^M(\vec{q}) G_M(Q^2).$$

Different current components and momenta give enough equations to determine the two Sachs form factors.

Then

- repeat for several Q^2
- fit $G_E(Q^2)$ near zero
- slope $\rightarrow$ charge radius

Maiani et al., NPB **293**, 420 (1987); Capitani et al., PRD **86**, 074502 (2012).

Quark-Line Disconnected Diagrams

Backup

The current can enter in two ways

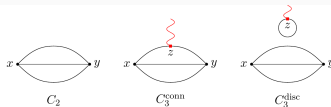
Connected

The current hits a quark line joining the proton source and sink.

Disconnected

The current creates a **sea-quark loop**, correlated with the proton only through the same gluon background.

Numerically noisy and expensive,
but required for precision.



C_2 , connected C_3 , and disconnected C_3 .

Djukanovic et al., PRL **132**, 211901 (2024) — includes disconnected diagrams in a precision nucleon-radius calculation.

Electromagnetic Form Factors: The Bridge to the Radius

Backup

Figure message: lattice QCD computes the same $G_E(Q^2)$ curve inferred from scattering; its low- Q^2 slope determines r_E^p .

Low- Q^2 slope

$$G_E(Q^2) = 1 - \frac{Q^2}{6} \langle r_E^2 \rangle + \dots$$

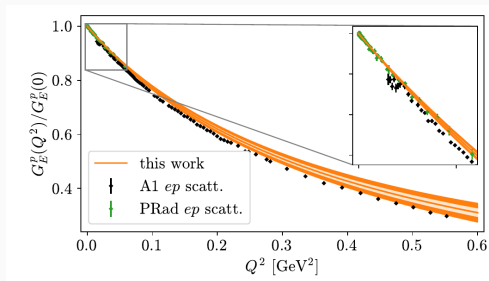
$$\langle r_E^2 \rangle = -6 \left. \frac{dG_E}{dQ^2} \right|_0$$

Lattice chain

$$C_2, C_3^\mu \rightarrow \langle p' | J^\mu | p \rangle$$

$$\rightarrow G_E(Q^2) \rightarrow r_E. \quad \text{Orange: lattice QCD; green: PRad; black: A1 Mainz ep data.}$$

Sachs, Phys. Rev. **126**, 2256 (1962); Djukanovic et al., PRL **132**, 211901 (2024).



Results

Mainz/CLS Proton Charge Radius

Final r_E^p after continuum and physical-point fits.

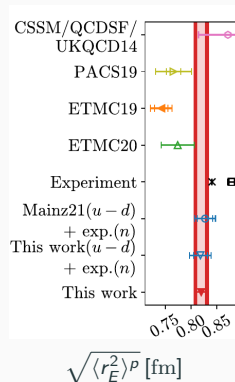
Backup

Djukanovic et al. (2024)

$$r_E^p = 0.820(14) \text{ fm}$$

**Compatible with the modern
small-radius picture.**

This is a first-principles QCD result after
controlled extrapolations.



Djukanovic et al., PRL 132, 211901 (2024) [arXiv:2309.07491].

Lattice vs. Experiment: Comparison

Lattice QCD

$$r_E^p = 0.820(14) \text{ fm}$$

PDG 2024

$$r_E^p = 0.8409(4) \text{ fm}$$

Compatible within about 1.5σ .

Lattice QCD supports the modern small-radius picture, but is not yet at PDG precision.

Lattice: Djukanovic et al., PRL **132**, 211901 (2024); double errors are stat/syst. Exp.: PDG 2024.

Summary and Outlook

Summary: What Did We Learn?

1. Experiments infer the radius.

cross sections $\rightarrow G_E(Q^2)$, energy levels $\rightarrow r_E^p$, $\langle r_E^2 \rangle = -6 G_E'(0)$.

2. Modern experiments favour the small-radius picture.

μH : 0.84184(67) fm, PDG 2024 : 0.8409(4) fm.

3. Lattice QCD computes the same proton response from QCD.

$$C_2, C_3^\mu \rightarrow \langle p' | J_{\text{em}}^\mu | p \rangle \rightarrow G_E(Q^2) \rightarrow \boxed{r_E^p = 0.820(14) \text{ fm}}.$$

The Mainz/CLS result is compatible with the modern picture within about 1.5σ .

Outlook: From Consistency to Precision

Lattice QCD

- more configurations / measurements $\Rightarrow$ reduce statistical errors
- finer- a ensembles $\Rightarrow$ controlled $a \rightarrow 0$ limit
(costly: critical slowing down)
- learned / ML samplers $\Rightarrow$ may accelerate gauge-field generation

Goal: first-principles QCD from consistency check to sub-percent precision.

Critical slowing: Schaefer et al., NPB **845**, 93 (2011). ML samplers: Albergo et al. (2019), Kanwar et al. (2020).

Thank you!

Questions?

Backup Roadmap

Puzzle & experiments

- Method tension
- Lamb shift size
- Zemach/Friar

Form factors

- ep formula map
- OPE amplitude
- Why Mott factor
- Feynman diagram
- Current form
- ep cross section
- Slope as radius
- G_M figure
- Why Sachs?

Lattice details

- Why non-perturbative?
- Why links?
- Why Euclidean?
- Proton operator
- Wick contraction
- Disconnected
- Flavor currents
- Fig. 3 labels
- Ratio method
- Gauge weight
- z -expansion
- Fermion actions

Why Could the Methods Disagree? (Backup)

Literature framing. Pohl et al. (2013) define the puzzle as the significant disagreement between μH spectroscopy and the radius extracted from electronic H / ep scattering. Later reviews describe the response as a program of checking theory, questioning experimental systematics, and designing new measurements.

Where could the tension enter?

- ep scattering: low- $Q^2 \rightarrow 0$ extrapolation, absolute normalisation, radiative corrections, two-photon exchange.
- H / μH spectroscopy: bound-state QED, $R_\infty - r_E^p$ correlation, proton polarizability / two-photon exchange.
- **More radical:** lepton-universality violation or muon-specific new interactions; interesting but highly constrained.

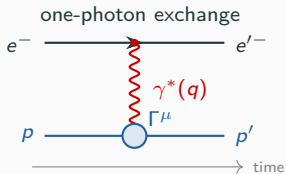
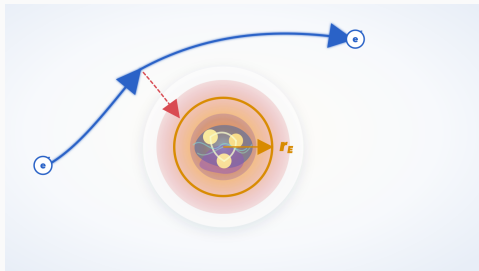
How did people try to improve it?

- **PRad:** very low Q^2 , windowless target, magnetic-spectrometer-free normalisation.
- **New H spectroscopy:** different transitions to separate R_∞ and r_E^p ; stress-test bound-state QED.
- **MUSE, MAGIX, PRad-II:** new e/μ scattering and lower Q^2 data to isolate systematics.

Modern wording: refined ep and H measurements moved toward the μH value, but the issue is best viewed as a precision-consistency test of scattering, spectroscopy, QED theory, and possible lepton universality.

Pohl et al., Ann. Rev. Nucl. Part. Sci. **63**, 175 (2013); Karr, Marchand, Voutier, Nat. Rev. Phys. **2**, 601 (2020); Xiong et al., Nature **575**, 147 (2019); Carlson & Freid, PRD **92**, 095024 (2015); Maisenbacher et al., Nature **650**, 845 (2026)

Elastic Scattering Formula Map (Backup)



$$\frac{d\sigma}{d\Omega} = \sigma_{\text{Mott}} S_p(Q^2, \theta)$$

$$\sigma_{\text{Mott}} = \frac{\alpha^2 \cos^2(\theta/2)}{4E^2 \sin^4(\theta/2)} \frac{E'}{E}$$

point-charge QED baseline

$$S_p = (G_E^2 + \tau G_M^2)/(1 + \tau) + 2\tau G_M^2 \tan^2 \frac{\theta}{2}$$

proton structure

$$J_{\text{em}}^\mu = \sum_q e_q \bar{q} \gamma^\mu q$$

EM current operator

$$\langle p' | J_{\text{em}}^\mu | p \rangle = \bar{u}(p') \left[\gamma^\mu F_1 + \frac{i\sigma^{\mu\nu} q_\nu}{2M_p} F_2 \right] u(p)$$

Dirac/Pauli form factors

$$G_E = F_1 - \tau F_2, \quad G_M = F_1 + F_2$$

Sachs form factors

$$\langle r_E^2 \rangle = -6 \left. \frac{dG_E}{dQ^2} \right|_{Q^2=0}$$

charge radius

Hofstadter, Rev. Mod. Phys. **28**, 214 (1956); Bernauer et al. (A1), PRL **105**, 242001 (2010); Gao & Vanderhaeghen, RMP **94**, 015002 (2022).

Where Does the Factorised Amplitude Come From? (Backup)

Start from the path integral / S-matrix expansion Local $U(1)_{\text{em}}$ gauge invariance gives minimal coupling: the photon field A_μ couples to the conserved electromagnetic current.

$$Z = \int [D\Phi] \exp \left\{ iS_0 + i \int d^4x \mathcal{L}_{\text{int}}(x) \right\}, \quad \mathcal{L}_{\text{int}} = -e A_\mu j_e^\mu - e A_\mu J_{\text{em}}^\mu.$$

$$j_e^\mu = \bar{e} \gamma^\mu e, \quad J_{\text{em}}^\mu = \sum_q Q_q \bar{q} \gamma^\mu q, \quad Q_u = \frac{2}{3}, \quad Q_d = Q_s = -\frac{1}{3}.$$

$$e^i \int \mathcal{L}_{\text{int}} = 1 + i \int d^4x \mathcal{L}_{\text{int}}(x) + \frac{i^2}{2} \int d^4x d^4y T\{\mathcal{L}_{\text{int}}(x) \mathcal{L}_{\text{int}}(y)\} + \dots$$

Lowest connected $ep \rightarrow ep$ term

$$\frac{i^2}{2} 2e^2 \int d^4x d^4y T\{A_\mu(x) j_e^\mu(x) A_\nu(y) J_{\text{em}}^\nu(y)\}.$$

$\langle T A_\mu(x) A_\nu(y) \rangle_0 = D_{\mu\nu}(x-y) \implies$ one photon line between the two currents.

$$\langle e(k') | j_e^\mu | e(k) \rangle = \underbrace{\bar{u}(k') \gamma^\mu u(k)}_{\ell^\mu \text{ known}} \quad \langle p' | J_{\text{em}}^\nu | p \rangle = \underbrace{H^\nu}_{\text{proton response}}$$

$$i\mathcal{M} = (-ie)^2 \ell_\mu \frac{-ig^{\mu\nu}}{q^2} H_\nu \implies \mathcal{M} \simeq \frac{e^2}{Q^2} \underbrace{\ell_\mu}_{\text{known}} \underbrace{H^\mu}_{\text{proton}}, \quad Q^2 = -q^2.$$

Why a Mott Factor Times Proton Structure? (Backup)

One-photon exchange separates the known lepton current from the proton current

$$\mathcal{M} = \frac{e^2}{Q^2} \underbrace{\bar{u}(k')\gamma_\mu u(k)}_{\ell_\mu \text{ known}} \underbrace{\langle p' | J_{\text{em}}^\mu | p \rangle}_{H^\mu \text{ proton}}.$$

$$|\overline{\mathcal{M}}|^2 = \frac{e^4}{Q^4} L_{\mu\nu} W^{\mu\nu}, \quad L_{\mu\nu} = \ell_\mu \ell_\nu^*, \quad W^{\mu\nu} = H^\mu H^{\nu*}.$$

Point-charge baseline

Real proton response

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{point}} \equiv \sigma_{\text{Mott}}.$$

$$W_{\text{elastic}}^{\mu\nu} \implies G_E(Q^2), G_M(Q^2).$$

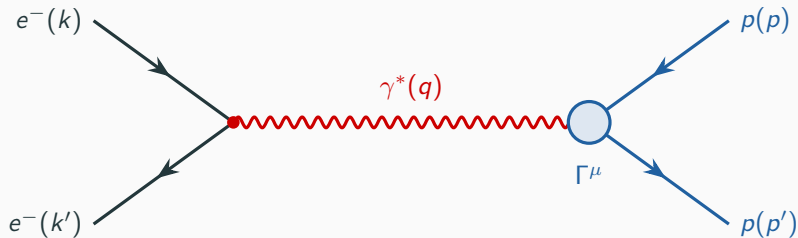
known QED lepton current, photon propagator, and elastic kinematics.

covariance, current conservation, and spin- $\frac{1}{2}$ leave two elastic responses.

$$\frac{d\sigma}{d\Omega} = \underbrace{\sigma_{\text{Mott}}}_{\text{known point-charge QED}} \underbrace{S_p(Q^2, \theta)}_{\text{dimensionless proton structure}}.$$

This is a convenient factorisation of the one-photon-exchange result, not an extra dynamical assumption.

Form Factors from the Feynman Diagram (Backup)



$$\langle p' | J_{\text{em}}^\mu(0) | p \rangle = \bar{u}(p') \left[\gamma^\mu F_1(Q^2) + \frac{i\sigma^{\mu\nu} q_\nu}{2M_N} F_2(Q^2) \right] u(p), \quad Q^2 = -q^2.$$

Peskin & Schroeder, *An Introduction to Quantum Field Theory*, Ch. 6; Sachs, Phys. Rev. **126**, 2256 (1962).

Why This Proton Current Form? I (Backup)

Start from the unknown proton vertex

$$\langle p' | J_{\text{em}}^\mu(0) | p \rangle = \bar{u}(p') \Gamma^\mu(P, q) u(p), \quad P = p' + p, \quad q = p' - p, \quad Q^2 = -q^2.$$

1. Lorentz covariance

$$\Gamma^\mu = a_1 \gamma^\mu + a_2 \frac{P^\mu}{2M_p} + a_3 \frac{q^\mu}{2M_p} + a_4 \frac{i\sigma^{\mu\nu} q_\nu}{2M_p}.$$

- Γ^μ must transform as a vector.
- $a_i = a_i(Q^2)$: the only invariant left is q^2 .
- $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$ carries spin; $\sigma^{\mu\nu} q_\nu$ gives the Pauli/magnetic response.
- Parity forbids axial structures such as $\gamma^\mu \gamma_5$.

Why This Proton Current Form? II (Backup)

2. Current conservation

$$q_\mu \langle p' | J_{\text{em}}^\mu | p \rangle = 0.$$

$$q \cdot P = p'^2 - p^2 = 0,$$

$$q_\mu \sigma^{\mu\nu} q_\nu = 0 \quad (\sigma^{\mu\nu} = -\sigma^{\nu\mu}).$$

$$\begin{aligned} \bar{u}(p') \gamma \cdot q u(p) &= \bar{u}(p') (\gamma \cdot p' - \gamma \cdot p) u(p) \\ &= 0. \end{aligned}$$

So the q^μ term is removed: $a_3 = 0$.

Why This Proton Current Form? III (Backup)

3. Gordon identity removes the P^μ basis

$$\bar{u}(p') \frac{P^\mu}{2M_p} u(p) = \bar{u}(p') \left[\gamma^\mu - \frac{i\sigma^{\mu\nu} q_\nu}{2M_p} \right] u(p).$$

Therefore the elastic proton current needs two scalar functions

$$\langle p' | J_{\text{em}}^\mu | p \rangle = \bar{u}(p') \left[\gamma^\mu F_1(Q^2) + \frac{i\sigma^{\mu\nu} q_\nu}{2M_p} F_2(Q^2) \right] u(p)$$

$$F_1^p(0) = 1, \quad F_2^p(0) = \kappa_p = 1.793; \quad F_1^n(0) = 0, \quad F_2^n(0) = \kappa_n = -1.913.$$

F_1 , F_2 are the two scalar functions containing QCD dynamics.

From One-Photon Exchange to the Radius I (Backup)

Insert the proton current into the one-photon exchange amplitude

$$i\mathcal{M} = (-ie)^2 \bar{u}_e(k') \gamma_\mu u_e(k) \frac{-ig^{\mu\nu}}{q^2} \bar{u}_p(p') \Gamma_\nu(q) u_p(p), \quad q = k - k' = p' - p.$$

$$\Gamma^\nu(q) = \gamma^\nu F_1(Q^2) + \frac{i\sigma^{\nu\alpha} q_\alpha}{2M_p} F_2(Q^2), \quad Q^2 = -q^2, \quad \tau = \frac{Q^2}{4M_p^2}.$$

Sachs variables

$$G_E = F_1 - \tau F_2,$$

$$G_M = F_1 + F_2,$$

$$\epsilon = \left[1 + 2(1 + \tau) \tan^2 \frac{\theta_e}{2} \right]^{-1}.$$

From One-Photon Exchange to the Radius II (Backup)

Mott baseline and reduced cross section

$$\sigma_{\text{Mott}} \equiv \left(\frac{d\sigma}{d\Omega} \right)_{\text{point}} = \frac{\alpha^2 \cos^2(\theta_e/2)}{4E^2 \sin^4(\theta_e/2)} \frac{E'}{E},$$

$$\sigma_R \equiv \frac{\epsilon(1+\tau)}{\sigma_{\text{Mott}}} \frac{d\sigma}{d\Omega} = \epsilon G_E^2 + \tau G_M^2.$$

Rosenbluth separation

$$\text{fixed } Q^2: \quad \sigma_R(\epsilon) = \underbrace{G_E^2}_{\text{slope}} \epsilon + \underbrace{\tau G_M^2}_{\text{intercept}}.$$

Vary E_{beam} and θ_e to scan ϵ at fixed Q^2 .

Charge radius

$$G_E(Q^2) = 1 - \frac{1}{6} \langle r_E^2 \rangle Q^2 + \mathcal{O}(Q^4),$$

$$\langle r_E^2 \rangle = -6 \left. \frac{dG_E}{dQ^2} \right|_{Q^2=0}.$$

Why the G_E Slope Is a Radius (Backup)

Breit-frame intuition

$$G_E(Q^2) \simeq \int d^3r \rho_E(\vec{r}) e^{i\vec{q} \cdot \vec{r}}, \quad \int d^3r \rho_E(\vec{r}) = 1.$$

- $Q^2 = \vec{q}^2$ in the Breit frame.
- $G_E(0) = 1$: total proton charge.
- Larger charge radius $\Rightarrow$ faster falloff with Q^2 .

Small- q expansion

$$e^{i\vec{q} \cdot \vec{r}} = 1 + i\vec{q} \cdot \vec{r} - \frac{1}{2}(\vec{q} \cdot \vec{r})^2 + \dots$$

$$\int \rho_E \vec{q} \cdot \vec{r} = 0, \quad \int \rho_E (\vec{q} \cdot \vec{r})^2 = \frac{\vec{q}^2}{3} \langle r_E^2 \rangle.$$

$1/6 = (1/2) \times (1/3)$: Taylor expansion $\times$ isotropic average.

$$G_E(Q^2) = 1 - \frac{Q^2}{6} \langle r_E^2 \rangle + \mathcal{O}(Q^4).$$

$$\boxed{\langle r_E^2 \rangle = -6 \left. \frac{dG_E}{dQ^2} \right|_{Q^2=0}}$$

This is the nonrelativistic/Breit-frame picture; the exact relativistic definition is the current matrix element.

Magnetic Form Factor G_M (Backup)

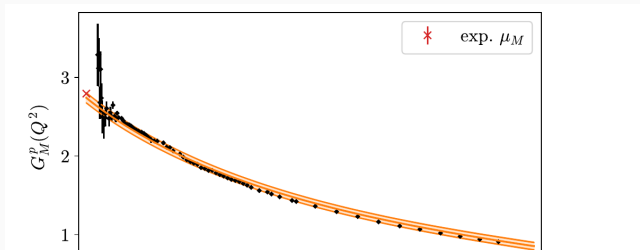
Same lattice calculation also gives G_M , but it is not needed for the charge radius.

Magnetic Sachs form factor

$$G_M = F_1 + F_2.$$

$$\langle r_M^2 \rangle = -\frac{6}{G_M(0)} \left. \frac{dG_M}{dQ^2} \right|_{Q^2=0}.$$

- G_E : charge response $\rightarrow r_E$
- G_M : magnetic response $\rightarrow r_M$
- Main talk follows G_E only



Djukanovic et al. (2024), PRL Fig. 2: $G_M^p(Q^2)$ panel.

Why Sachs G_E , G_M ? I (Backup)

Step 1: F_1, F_2 come from symmetry, not from “electric vs magnetic”

$$\langle p' | J_{\text{em}}^\mu | p \rangle = \bar{u}(p') \left[\gamma^\mu F_1(Q^2) + \frac{i\sigma^{\mu\nu} q_\nu}{2M_p} F_2(Q^2) \right] u(p).$$

- This is the most general spin- $\frac{1}{2}$ current allowed by Lorentz covariance and current conservation.
- F_1 and F_2 are therefore **covariant form factors**.
- At finite Q^2 , they do not separately equal “charge distribution” and “magnetisation distribution”.

Step 2: choose a frame where charge and current are easy to read

$$\text{Breit frame: } q^\mu = (0, \mathbf{q}), \quad p^\mu = (E, -\mathbf{q}/2), \quad p'^\mu = (E, \mathbf{q}/2), \quad Q^2 = \mathbf{q}^2.$$

- No energy transfer: $q^0 = 0$.
- J^0 behaves like a charge-density response.
- $\mathbf{J}$ behaves like a current/magnetic response.

Why Sachs G_E , G_M ? II (Backup)

Step 3: in the Breit frame, the current components pick special combinations

$$\langle p' | J_{\text{em}}^0 | p \rangle_{\text{B}} \propto \chi'^{\dagger} \chi \underbrace{(F_1 - \tau F_2)}_{G_E(Q^2)}, \quad \tau = \frac{Q^2}{4M_p^2}.$$

$$\langle p' | \mathbf{J}_{\text{em}} | p \rangle_{\text{B}} \propto i \chi'^{\dagger} \frac{\boldsymbol{\sigma} \times \mathbf{q}}{2M_p} \chi \underbrace{(F_1 + F_2)}_{G_M(Q^2)}.$$

$$\boxed{G_E = F_1 - \tau F_2},$$

$$\boxed{G_M = F_1 + F_2}.$$

Why “electric”?

Why “magnetic”?

$$J^0 \longrightarrow G_E(Q^2)$$

$$\mathbf{J} \longrightarrow G_M(Q^2)$$

$$G_E(Q^2) \sim \int d^3r e^{i\mathbf{q}\cdot\mathbf{r}} \rho_E(\mathbf{r}) \quad (\text{nonrelativistic picture}).$$

$G_M(0) = \mu_p$, spatial current encodes magnetisation.

$$\langle r_E^2 \rangle = -6 \left. \frac{dG_E}{dQ^2} \right|_{Q^2=0}.$$

$$\langle r_M^2 \rangle = -\frac{6}{G_M(0)} \left. \frac{dG_M}{dQ^2} \right|_{Q^2=0}.$$

Why $\Delta E_{\text{fs}} \propto |\psi(0)|^2 r_E^2$? (Backup)

Use the same G_E that scattering measures

Slope derivation

$$G_E(Q^2) = 1 - \frac{Q^2}{6} \langle r_E^2 \rangle + \mathcal{O}(Q^4).$$

$$V(\mathbf{q}) = -\frac{4\pi Z\alpha}{\mathbf{q}^2} G_E(\mathbf{q}^2) = -\frac{4\pi Z\alpha}{\mathbf{q}^2} + \frac{2\pi Z\alpha}{3} \langle r_E^2 \rangle + \dots$$

$$\delta V(\mathbf{r}) = \frac{2\pi Z\alpha}{3} \langle r_E^2 \rangle \delta^{(3)}(\mathbf{r}).$$

First-order perturbation theory

So for S states

$$\Delta E_{n\ell}^{\text{fs}} = \langle n\ell m | \delta V | n\ell m \rangle = \frac{2\pi Z\alpha}{3} \langle r_E^2 \rangle |\psi_{n\ell m}(0)|^2.$$

$$\Delta E_{nS}^{\text{fs}} = \frac{2}{3n^3} (Z\alpha)^4 m_r^3 \langle r_E^2 \rangle$$

$$|\psi_{nS}(0)|^2 = \frac{(Z\alpha m_r)^3}{\pi n^3}, \quad |\psi_{nP}(0)|^2 = 0.$$

- $2S$ shifts because it samples the origin.
- $2P$ hardly shifts because $\psi_{2P}(0) = 0$.
- Measured $2S-2P$: QED + finite size; fit r_E .

Eides, Grotch & Shelyuto, Phys. Rep. **342**, 63 (2001); Antognini et al., Science **339**, 417 (2013).

Perturbation vs. Non-Perturbative QCD & Lattice (Backup)

Perturbation theory needs a small parameter

$$\mathcal{O} = \mathcal{O}_0 + \alpha_s \mathcal{O}_1 + \alpha_s^2 \mathcal{O}_2 + \dots$$

$\alpha_s \ll 1 \Rightarrow$ higher orders are suppressed.

Then a few Feynman diagrams can give a controlled approximation.

At proton scales this hierarchy is lost

$$r \sim 1 \text{ fm}, \quad Q \sim \Lambda_{\text{QCD}} \sim \text{few} \times 100 \text{ MeV}.$$

$\alpha_s(Q) \sim \mathcal{O}(1) \Rightarrow 1, \alpha_s, \alpha_s^2, \dots$ are not ordered.

Many gluon exchanges, sea-quark fluctuations, and confinement are part of the proton structure, not small corrections.

Therefore the desired current matrix element is non-perturbative

$\langle p' | J_{\text{em}}^\mu | p \rangle$ must be computed from QCD beyond a low-order diagram expansion.

$$\langle \mathcal{O} \rangle = \frac{1}{Z} \int [DU][D\bar{q}][Dq] \mathcal{O}[U, q, \bar{q}] e^{-S_{\text{QCD}}} \xrightarrow[\text{MC}]{\text{lattice}} \text{controlled } a \rightarrow 0, L \rightarrow \infty \text{ result.}$$

Lepage, “Lattice QCD for Novices,” arXiv:hep-lat/0506036 (2005); Wittig, “QCD on the Lattice,” Springer (2020), Sec. 5.2.

Why Do Gluons Live on Links? (Backup)

Quarks live at spacetime points

$$\psi(x) \rightarrow \Omega(x) \psi(x), \quad \bar{\psi}(x) \rightarrow \bar{\psi}(x) \Omega^\dagger(x).$$

- $\Omega(x) \in SU(3)$ is a **local colour rotation**.
- Neighbouring sites have independent colour frames: $\Omega(x) \neq \Omega(x + a\hat{\mu})$.
- A naive hopping term is not gauge invariant:

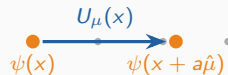
$$\bar{\psi}(x) \psi(x + a\hat{\mu}) \rightarrow \bar{\psi}(x) \Omega^\dagger(x) \Omega(x + a\hat{\mu}) \psi(x + a\hat{\mu}).$$

The gluon is the connection between sites

$$U_\mu(x) \simeq \exp[iag A_\mu(x + a\hat{\mu}/2)] \in SU(3).$$

$$U_\mu(x) \rightarrow \Omega(x) U_\mu(x) \Omega^\dagger(x + a\hat{\mu}).$$

$$\bar{\psi}(x) U_\mu(x) \psi(x + a\hat{\mu}) \quad \text{is gauge invariant.}$$



Gauge action: the smallest closed transporter is the plaquette

$$U_{\mu\nu}(x) = U_\mu(x) U_\nu(x + a\hat{\mu}) U_\mu^\dagger(x + a\hat{\nu}) U_\nu^\dagger(x) \simeq \exp[ia^2 g F_{\mu\nu}(x)].$$

The Wilson plaquette combination obeys

$$1 - \frac{1}{3} \text{Re Tr } U_{\mu\nu} \propto a^4 \text{Tr } F_{\mu\nu} F_{\mu\nu} + \mathcal{O}(a^6),$$

which gives the Euclidean gauge action in the continuum limit.

Why Euclidean, Not Directly Minkowski? (Backup)

Minkowski path integral

$$Z_M = \int [D\Phi] e^{iS_M[\Phi]}.$$

- e^{iS_M} is a rapidly oscillating phase.
- It is not a positive probability weight.
- Configurations cancel against each other $\Rightarrow$ **severe sign problem**.

Wick rotation

$$t = -i\tau, \quad e^{iS_M} \longrightarrow e^{-S_E}.$$

Euclidean path integral

$$Z_E = \int [D\Phi] e^{-S_E[\Phi]}.$$

- e^{-S_E} is a Boltzmann-like weight.
- After integrating out the fermions, zero-density QCD can be **sampled by Monte Carlo**.
- Euclidean time also filters **hadron ground states**:

$$C_2(\tau) = \sum_n |A_n|^2 e^{-E_n \tau} \xrightarrow[\tau \text{ large}]{} |A_0|^2 e^{-M_N \tau}.$$

For this talk: the charge radius uses spacelike form factors $G_E(Q^2)$, $Q^2 > 0$, which are naturally accessed from **Euclidean correlation functions** and then extrapolated to $Q^2 = 0$.

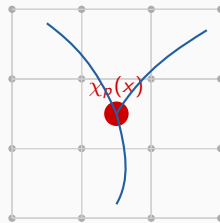
How Is a Proton Put on the Lattice? I (Backup)

Important point: we do **not** insert a little proton as a particle. The lattice path integral contains quark and gluon fields. A “proton” is created by a **composite operator** with the proton’s quantum numbers.

Example: proton interpolating operator

$$\chi_p^\alpha(x) = \epsilon^{abc} (u^{aT}(x) C \gamma_5 d^b(x)) u^{c\alpha}(x).$$

- uud : proton flavour content.
- ϵ^{abc} : colour singlet.
- $C\gamma_5$: spin-zero diquark structure; parity is selected with a projector.
- α : remaining Dirac spin index.



local product of three
quark fields at one site

$$\bar{\chi}_p(0) |0\rangle = Z_p |p\rangle + Z_{N^*} |N^*\rangle + Z_{N\pi} |N\pi\rangle + \dots$$

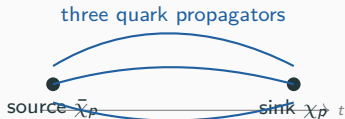
Meaning: the operator creates every state with proton quantum numbers; Euclidean time later filters out the **lightest one, the proton**.

Ioffe, Nucl. Phys. B **188**, 317 (1981); Wittig, “QCD on the Lattice”, Springer (2020), §5.2.

How Is a Proton Put on the Lattice? II (Backup)

Source and sink. Put $\bar{\chi}_p$ at the source time and χ_p at the sink time; then average over the QCD ensemble.

$$C_2(t) = \sum_{\vec{x}} \langle \chi_p(\vec{x}, t) \bar{\chi}_p(\vec{0}, 0) \rangle.$$
$$C_3^\mu(t, \tau, \vec{q}) = \sum_{\vec{x}, \vec{y}} e^{i\vec{q} \cdot \vec{y}} \langle \chi_p(\vec{x}, t) J_{\text{em}}^\mu(\vec{y}, \tau) \bar{\chi}_p(\vec{0}, 0) \rangle.$$



C_2 : create and
identify a proton



C_3 : insert current
and measure response

After integrating out the Grassmann quark fields, these pictures become products of quark propagators $S_q(x, 0) = D_q^{-1}(x, 0)$ on each gauge configuration. Large Euclidean separations isolate the ground-state term; an appropriate C_3/C_2 ratio then removes overlap and propagation factors.

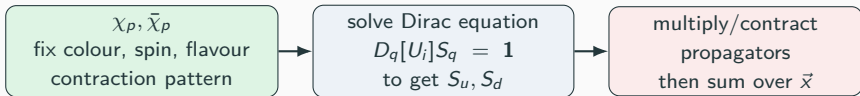
After Wick Contraction: What Is Computed? (Backup)

Before contraction:

$$C_2(t) = \sum_{\vec{x}} \langle \chi_p(\vec{x}, t) \bar{\chi}_p(\vec{0}, 0) \rangle, \quad \chi_p \sim \epsilon^{abc} (u^{aT} C \gamma_5 d^b) u^c.$$

Fixed gauge configuration U_i : Wick contracting the quark fields gives

$$\langle q(x) \bar{q}(y) \rangle_{U_i} = S_q(x, y; U_i) = D_q[U_i]^{-1}(x, y).$$



Schematic contracted form:

$$C_2(t; U_i) = \sum_{\vec{x}} c_{\chi_p} \left[S_u(x, 0; U_i), S_d(x, 0; U_i), S_u(x, 0; U_i) \right],$$

$$C_2(t) \simeq \frac{1}{N_{\text{cfg}}} \sum_{i=1}^{N_{\text{cfg}}} C_2(t; U_i).$$

$$C_{3,\text{conn}}^\mu : S_q(x, 0; U_i) \rightarrow S_q(x, y; U_i) \Gamma_q^\mu S_q(y, 0; U_i) \Rightarrow \text{appropriate ratio of } C_3^\mu \text{ and } C_2 \rightarrow \langle p' | J_{\text{em}}^\mu | p \rangle.$$

Ratio and Summation Methods (Backup)

Ratio method: choose a combination where source/sink overlap factors and leading Euclidean exponentials cancel,

$$R_\mu(\vec{q}, t, \tau) = \frac{C_3^\mu(\vec{q}, t, \tau)}{C_2(\vec{0}, t)} \sqrt{\frac{C_2(\vec{q}, t - \tau) C_2(\vec{0}, \tau) C_2(\vec{0}, t)}{C_2(\vec{0}, t - \tau) C_2(\vec{q}, \tau) C_2(\vec{q}, t)}} \xrightarrow{0 \ll \tau \ll t} \Pi_\mu.$$

$$\Pi_\mu(\vec{q}; \Gamma) = K_\mu^E(\vec{q}, \Gamma) G_E(Q^2) + K_\mu^M(\vec{q}, \Gamma) G_M(Q^2).$$

Known kinematics K_μ : energies, momentum, state normalization, and spin projector Γ .

Summation method: reduce excited-state contamination by summing over current insertion time,

$$S_\mu(\vec{q}, t) = \sum_{\tau=a}^{t-a} R_\mu(\vec{q}, t, \tau) \xrightarrow{t \text{ large}} c + t \Pi_\mu + \mathcal{O}(e^{-\Delta E t}).$$

Signal-to-noise

- baryon noise worsens exponentially with t
- use large statistics, smearing, multiple source-sink separations

Excited states

- N^* , $N\pi$, $N\pi\pi \dots$ also contribute
- use summation, multi-state fits, or GEVP

Maiani et al., NPB **293**, 420 (1987); Capitani et al., PRD **86**, 074502 (2012); Lepage (1989).

Why Disconnected Diagrams Matter (Backup)

Full QCD current insertion has two topologies.

$$J_{\text{em}}^\mu = \sum_q e_q \bar{q} \gamma^\mu q \Rightarrow C_3^{\text{conn}} + C_3^{\text{disc}}.$$

Connected

- J_{em}^μ hits a quark line joining source and sink.
- Physical picture: direct response of the proton quark lines.

Disconnected

- J_{em}^μ hits a **sea-quark loop**.
- Not connected by a quark line, but correlated through the same **gluon background U** .
- This is a genuine **sea-quark fluctuation**, not an IR divergence.

What is computed on each gauge field U_i ?

$$L_q^\mu(t; U_i) = \sum_{\vec{x}} \text{Tr}_{c,s} \left[\gamma^\mu D_q^{-1}(x, x; U_i) \right]$$

$$C_{3,\text{disc}}^\mu(t, \tau) \sim \left\langle C_2(t; U) L_q^\mu(\tau; U) \right\rangle_U - \left\langle C_2(t; U) \right\rangle_U \left\langle L_q^\mu(\tau; U) \right\rangle_U.$$

- $D_q^{-1}(x, x)$: all-to-all quark propagation back to the same point.
- Trace over colour and spin $\Rightarrow$ stochastic estimators.
- Numerically noisy and expensive, but required for precision form factors.

Djukanovic et al., PRD **109**, 094510 (2024); Djukanovic et al., PRL **132**, 211901 (2024).

Flavor Currents: What Are $u - d$ and $u + d$? (Backup)

Start from quark-flavor vector currents

$$J_u^\mu = \bar{u}\gamma^\mu u, \quad J_d^\mu = \bar{d}\gamma^\mu d, \quad J_s^\mu = \bar{s}\gamma^\mu s.$$

Isovector: $u - d$

$$J_{u-d}^\mu = J_u^\mu - J_d^\mu.$$

- Measures the proton–neutron difference:

$$G^{u-d} = G^p - G^n.$$

- In the isospin limit $m_u = m_d$, disconnected loops cancel:

$$L_u^\mu - L_d^\mu = 0.$$

- Hence $u - d$ is the cleanest lattice channel.

Isoscalar / octet: $u + d, u + d - 2s$

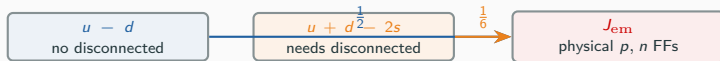
$$J_{u+d}^\mu = J_u^\mu + J_d^\mu, \quad J_8^\mu = J_u^\mu + J_d^\mu - 2J_s^\mu.$$

- Measures the proton–neutron common part.
- Disconnected loops add instead of cancel:

$$L_u^\mu + L_d^\mu \neq 0.$$

- Full $N_f = 2 + 1$ EM current uses the octet combination:

$$J_{\text{em}}^\mu = \frac{1}{2} J_{u-d}^\mu + \frac{1}{6} J_8^\mu.$$



How to Read $u - d$ and $+\text{exp.}(n)$ (Backup)

The labels in PRL Fig. 3 do not all mean the same level of calculation.

$u - d$

$+\text{exp.}(n)$

$$G^{u-d} = G^p - G^n.$$

$$G^p = G^{u-d} + G_{\text{exp}}^n \quad \text{schematically.}$$

- Pure isovector lattice result.
- Useful because disconnected diagrams cancel for $m_u = m_d$.
- It is **not yet a full proton EM result**; it gives the proton–neutron difference.
- Combine lattice $u - d$ with experimental neutron input.
- Good comparison, but not a fully ab initio proton result.
- For charge radii, the slope combination gives schematically

$$\langle r_E^2 \rangle^p = \langle r_E^2 \rangle^{u-d} + \langle r_E^2 \rangle_{\text{exp}}^n.$$

Speaking sentence: $u - d$ is clean but incomplete; $+\text{exp.}(n)$ uses neutron data to reconstruct a proton number.

PRL Fig. 3 caption: Mainz21 combines isovector results with PDG neutron values.

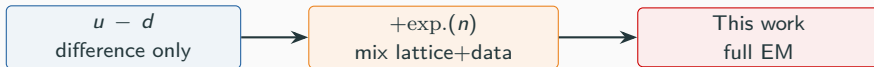
How to Read “This Work” in PRL Fig. 3 (Backup)

This work

$$J_{\text{em}}^{\mu} = \frac{1}{2}(u - d) + \frac{1}{6}(u + d - 2s).$$

- $u - d$ part: connected isovector channel.
- $u + d - 2s$ part: isoscalar/octet channel.
- Disconnected sea-quark loops are included instead of assumed away.
- Continuum, finite-volume, and physical-point extrapolations are controlled.
- Therefore this is the final first-principles proton result:

$$\sqrt{\langle r_E^2 \rangle^p} = 0.820(14) \text{ fm.}$$



PRL Fig. 3 caption: only ETMC19 and this work include disconnected contributions; Djukanovic et al., PRL 132, 211901 (2024).

The z -Expansion: Analytic Radius Extraction (Backup)

Problem: lattice data at discrete $Q^2 > 0$; need $dG_E/dQ^2|_{Q^2=0}$.

Traditional: dipole $G_E(Q^2) = (1 + Q^2/M_D^2)^{-2}$ — **model-dependent**.

z -expansion (analyticity-based parametrisation):

1. Conformal map $Q^2 \rightarrow z$:
$$z = \frac{\sqrt{t_{\text{cut}} + Q^2} - \sqrt{t_{\text{cut}} - t_0}}{\sqrt{t_{\text{cut}} + Q^2} + \sqrt{t_{\text{cut}} - t_0}}$$

 $t_{\text{cut}} = 4m_\pi^2$ (isovector) or $9m_\pi^2$ (isoscalar); t_0 : free param, optimises convergence
2. $|z| < 1$ guaranteed $\rightarrow$ convergent Taylor series
3. Expand: $G_E(z) = \sum_{k=0}^{k_{\text{max}}} a_k z^k$; coefficient priors/bounds and truncation tests control higher orders

For the proton, $G_E^p(0) = 1$, and the radius is

$$\langle r_E^2 \rangle^p = -6 \left. \frac{dG_E^p}{dz} \frac{dz}{dQ^2} \right|_{Q^2=0}.$$

If $t_0 = 0$, then $z(0) = 0$ and only a_1 enters the derivative. In Djukanovic et al., the z -expansion is a cross-check; the quoted central result uses model-averaged direct $B_\chi\text{PT}$ fits.

Hill, Paz, PRD **82**, 113005 (2010); Hill, EPJ Web Conf. **137**, 01023 (2017).

Finite Volume: How Large L is Needed? (Backup)

Periodic spatial box: $\vec{x} \equiv \vec{x} + L\hat{e}_i$ gives mirror images and quantised momenta. For many stable one-hadron observables, leading finite-volume effects scale as $\mathcal{O}(e^{-m_\pi L})$.

- **Empirical heuristic:** often $m_\pi L \gtrsim 4$ for masses & single-hadron overlaps $\Rightarrow$ tail $\sim e^{-m_\pi L}$ small at chosen tolerance (still observable-dependent).
- The safe procedure is to vary L or use finite-volume formulae; there is no universal box-size criterion for every observable.
- CLS open temporal boundaries address topology freezing at fine a ; they do *not* remove spatial finite-volume effects.

Lüscher, Commun. Math. Phys. **104**, 177 (1986); Commun. Math. Phys. **105**, 153 (1986); Beane et al., Nucl. Phys. B793, 379 (2008).

RG and Symanzik Scaling (Backup)

RG continuum limit: changing a means retuning bare parameters $g_0(a)$, $m_q(a)$ along a **line of constant physics**. Long-distance dimensionless observables are held fixed while $a \rightarrow 0$.

$$\mu \frac{dg_R}{d\mu} = \beta(g_R), \quad \mu \sim a^{-1}, \quad r_E(a) = r_E^{\text{cont}} + c_a a^2 + \dots$$

Symanzik effective theory: a lattice observable is described by continuum QCD plus cutoff corrections organised in powers of a . After non-perturbative $O(a)$ improvement of the action and current, the leading fit ansatz used here is a^2 , up to logarithms and higher powers.

Therefore: values at several lattice spacings test the predicted scaling and determine the intercept at $a = 0$. This is a controlled continuum extrapolation, not a free-form curve fit.

Symanzik, Nucl. Phys. **B226**, 205 (1983); Bruno et al., JHEP **02**, 043 (2015).

Sea Quarks and Gauge Configurations (Backup)

Start with **quark** and **gluon** fields

$$Z = \int [DU] [D\bar{\psi} D\psi] \exp \left[-S_g[U; \beta] - \sum_f \bar{\psi}_f D_f[U; m_f] \psi_f \right].$$

Integrate Grassmann fermions analytically

$$Z = \int [DU] \underbrace{e^{-S_g[U; \beta]} \prod_f \det D_f[U; m_f]}_{P[U] \text{ up to normalization}}.$$

What is sampled?

- The generated configurations are $U_\mu(x)$, not $\psi(x)$.
- The probability of U includes **sea-quark effects**:
 $P[U] \propto e^{-S_g[U; \beta]} \prod_{f=u,d,s,\dots} \det D_f[U; m_f].$
- In HMC the determinant is represented by **pseudofermion fields**:
 $\det(D^\dagger D) \propto \int [D\phi^\dagger D\phi] e^{-\phi^\dagger (D^\dagger D)^{-1} \phi}.$

Valence quark lines

- After choosing U_i , solve a Dirac equation:
 $D_q[U_i; m_q] S_q[U_i; m_q] = \mathbf{1}.$
- $S_q = D_q^{-1}$ is the quark propagator used in C_2, C_3 .
- Fermions enter twice: **determinant for sea quarks**; **propagators for measured quark lines**.

Where Does m_π Enter the Gauge Weight? (Backup)

m_π is not put into the path integral directly. It is the pion mass produced by the chosen light-quark masses in the fermion determinant.

Example: 2 + 1-flavour gauge weight

$$P[U] \propto e^{-S_g[U; \beta]} \det D_I[U; m_l]^2 \\ \times \det D_s[U; m_s], \quad m_l \simeq m_u \simeq m_d.$$

- m_l changes the determinant, so it changes the ensemble.
- The generated U_i 's are distributed according to that choice of (β, m_l, m_s) .
- A lighter m_l produces a lighter pion.

Chiral intuition

$$m_\pi^2 \simeq 2B m_l \quad (m_l = (m_u + m_d)/2).$$

- Modern ensembles tune m_l so that $m_\pi \simeq m_\pi^{\text{phys}}$.
- Older/heavier ensembles extrapolate observables in m_π .
- For a unitary calculation, valence and sea masses match; otherwise it is partially quenched.

Doublers and the Wilson Operator (Backup)

Naive lattice fermions double the species

$$D_{\text{naive}}(p) = m + \frac{i}{a} \sum_{\mu} \gamma_{\mu} \sin(ap_{\mu}).$$

Since $\sin(ap_{\mu}) = 0$ at $p_{\mu} = 0$ and $p_{\mu} = \pi/a$, one flavour becomes $2^4 = 16$ light fermions in four dimensions.

Wilson's cure: add a momentum-dependent mass for the doublers,

$$D_W(p) = D_{\text{naive}}(p) + \frac{r}{a} \sum_{\mu} (1 - \cos(ap_{\mu})).$$

For $p_{\mu} \sim \pi/a$, this mass is $\mathcal{O}(1/a)$, so doublers decouple as $a \rightarrow 0$.

Coordinate-space Wilson operator

$$(D_W q)(x) = \left(m_0 + \frac{4r}{a}\right) q(x) - \frac{1}{2a} \sum_{\mu} \left[(r - \gamma_{\mu}) U_{\mu}(x) q(x + \hat{\mu}) + (r + \gamma_{\mu}) U_{\mu}^{\dagger}(x - \hat{\mu}) q(x - \hat{\mu}) \right].$$

$$D_{\text{SW}} = D_W + c_{\text{SW}} \frac{ia}{4} \sigma_{\mu\nu} \hat{F}_{\mu\nu} \quad (\text{clover improvement}).$$

- **Wilson term** removes doublers but breaks **chiral symmetry** at finite a .
- **Clover improvement** removes leading $\mathcal{O}(a)$ artefacts after tuning.
- Other fermion actions make different compromises; see next backup slide.

Common Fermion Discretisations (Backup)

The issue: doublers and chiral symmetry cannot both be handled for free. Different lattice fermions make different compromises.

Wilson / clover

- removes doublers with Wilson term
- chiral symmetry broken at $a > 0$
- clover term improves $\mathcal{O}(a)$ artefacts
- moderate cost; good for nucleon structure

Twisted-mass Wilson

- Wilson-type action at nonzero twist
- automatic $\mathcal{O}(a)$ improvement at maximal twist
- often used for nucleon EM observables

Staggered / HISQ

- cheap; remnant chiral symmetry
- has taste degrees of freedom
- common in HVP, thermodynamics, heavy flavour

Domain-wall / overlap

- near-exact / Ginsparg–Wilson chiral symmetry
- expensive
- useful for kaon physics and chiral observables

This talk: CLS $N_f = 2 + 1$ Wilson–clover ensembles — $\mathcal{O}(a)$ -improved, moderate cost, chiral symmetry restored as $a \rightarrow 0$.

Nielsen & Ninomiya (1981); Sheikholeslami & Wohlert (1985); Frezzotti & Rossi (2004); Follana et al. (2007); Kaplan (1992); Neuberger (1998).

Zemach and Friar Radii (Backup)

Beyond r_E , the **Friar radius** (r_F) enters higher-order finite-size corrections to the Lamb shift, while the **Zemach radius** (r_Z) enters hyperfine splitting.

($\kappa_p = \mu_p/\mu_N - 1 \approx 1.793$: proton anomalous magnetic moment)

Definitions (convolutions of form factors):

$$r_Z = -\frac{4}{\pi} \int_0^\infty \frac{dQ}{Q^2} \left[\frac{G_E G_M}{1 + \kappa_p} - 1 \right], \quad r_F^3 = \frac{48}{\pi} \int_0^\infty \frac{dQ}{Q^4} \left[G_E^2 - 1 + \frac{1}{3} \langle r_E^2 \rangle Q^2 \right]$$

First lattice QCD calculation (Mainz group):

$$r_Z^p = 1.013(10)(12) \text{ fm}, \quad r_F^p = 1.301(12)(14) \text{ fm}$$

- r_F : higher-order charge-distribution correction in muonic atoms
- r_Z : leading elastic proton-structure correction to hyperfine splitting
- Compatible with most phenomenological determinations
- Computed from the same CLS ensembles and analysis framework

Djukanovic, von Hippel, Meyer, Ottnad, Salg, Wittig, PRD **110**, L011503 (2024) [arXiv:2309.17232].

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